

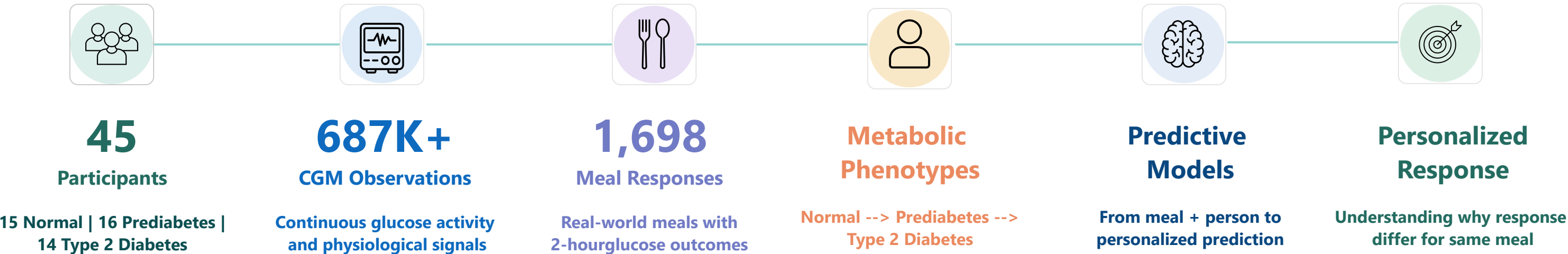
Why Similar Meals Create Different Glucose Responses

A multimodal analysis of CGM, nutrition, metabolic health, activity, gut health, and microbiome data



Total Participants
45
15 Normal | 16 Prediabetes | 14 T2D

Our Analytical Journey





15
Normal
Lower average glucose and variability

33.3%



16
Prediabetes
Intermediate but highly heterogeneous

35.6%



14
Type 2 Diabetes
Higher glucose and slower recovery

31.1%



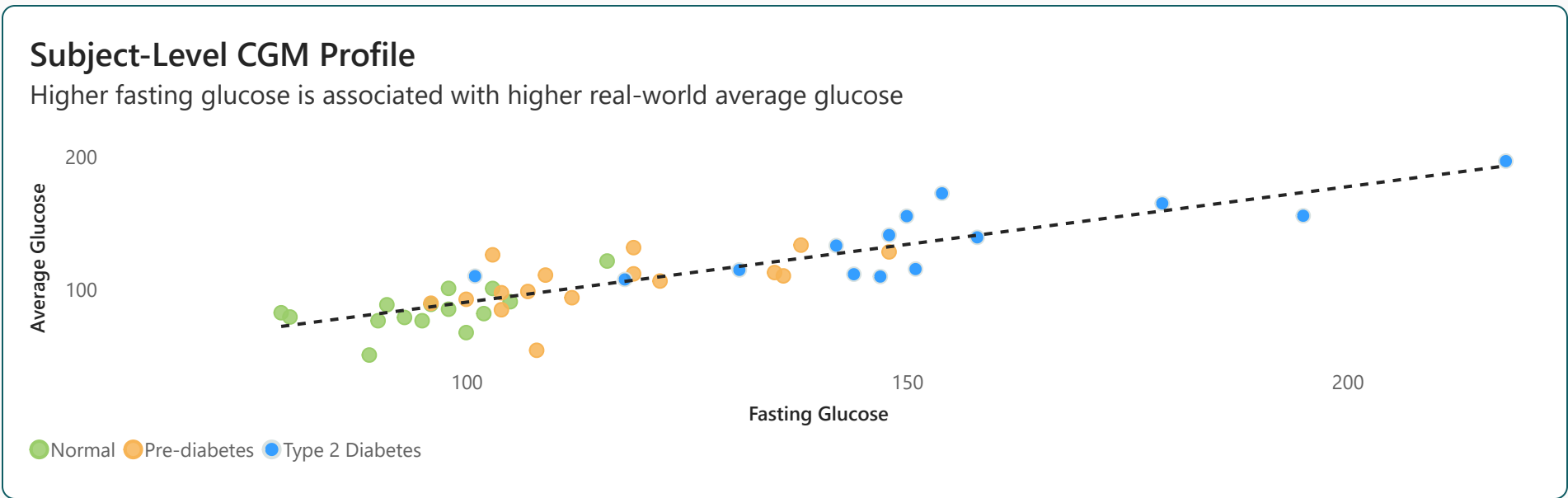
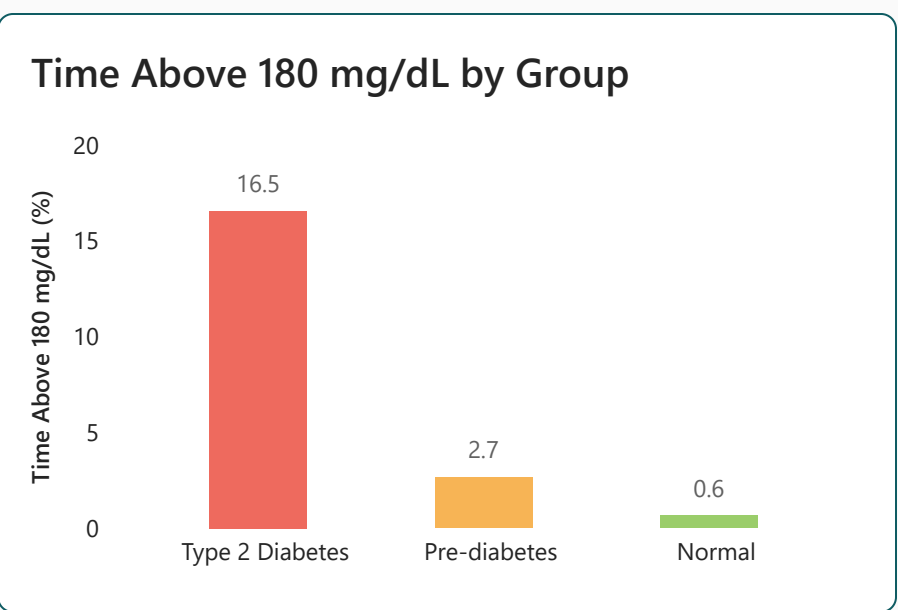
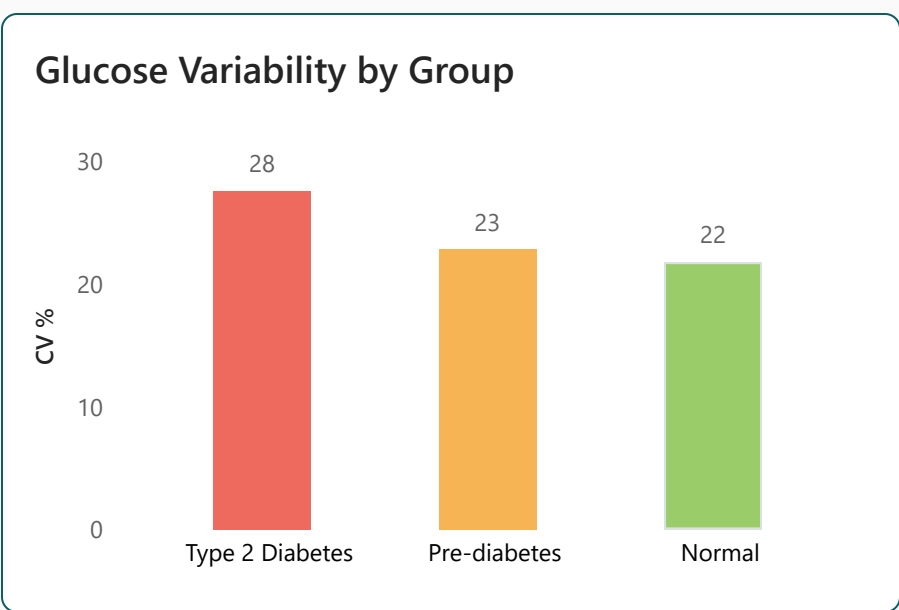
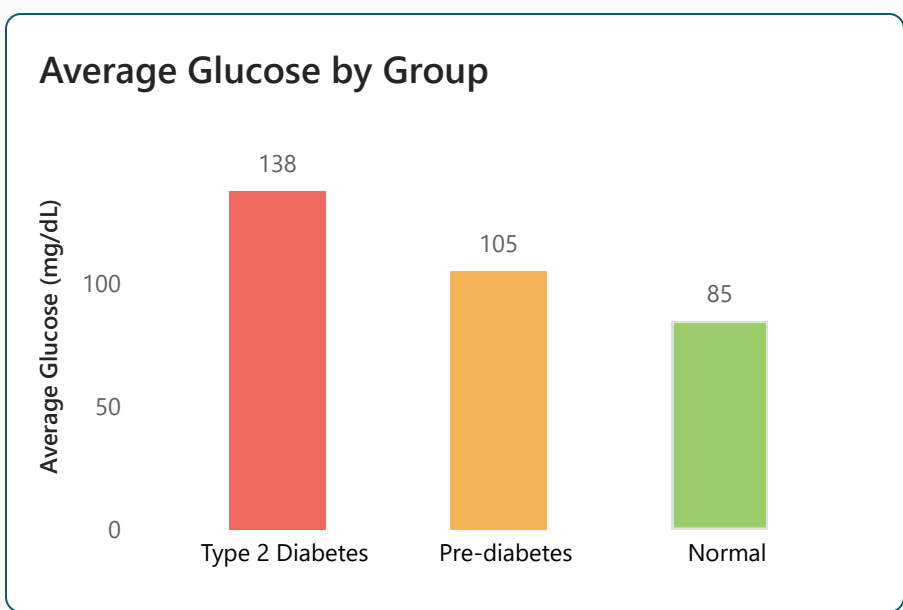
If two people eat nutritionally similar meals, why can one have a small glucose response while another experiences a major spike?



Goal: Discover how metabolic status, meal composition, activity, and biology interact to shape individual glucose responses.

CGM Progression Across Metabolic Groups

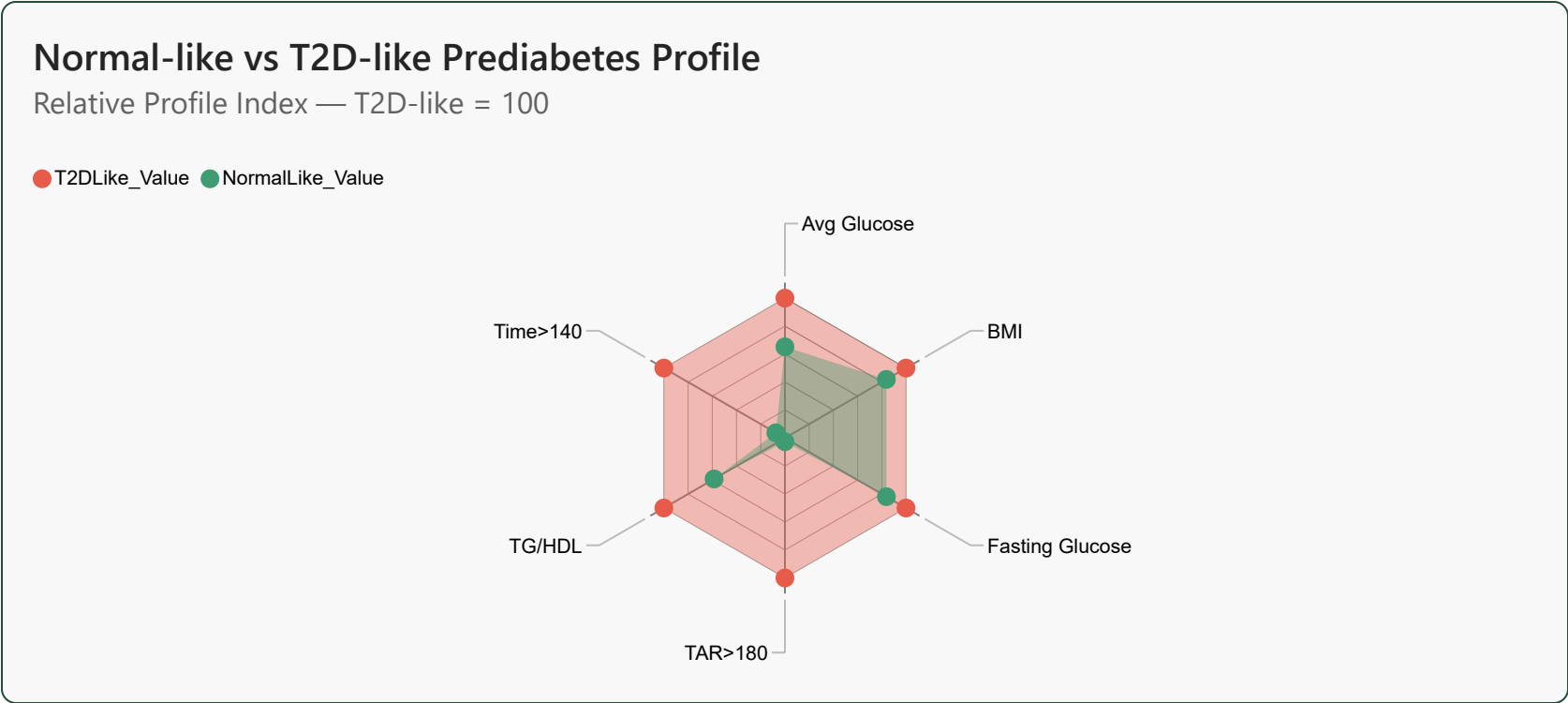
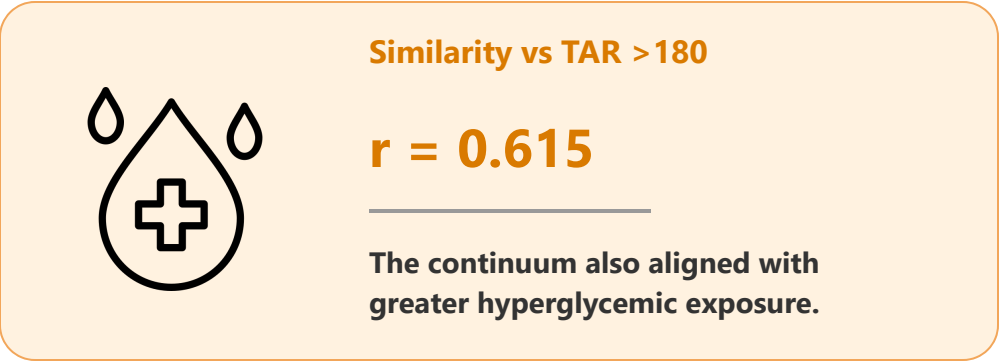
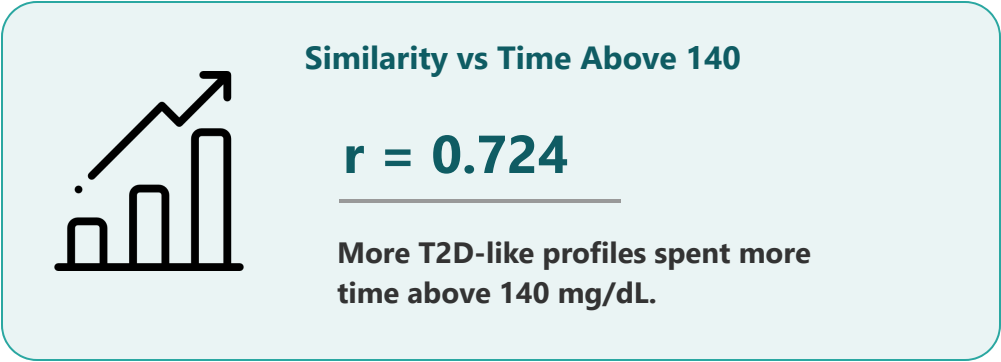
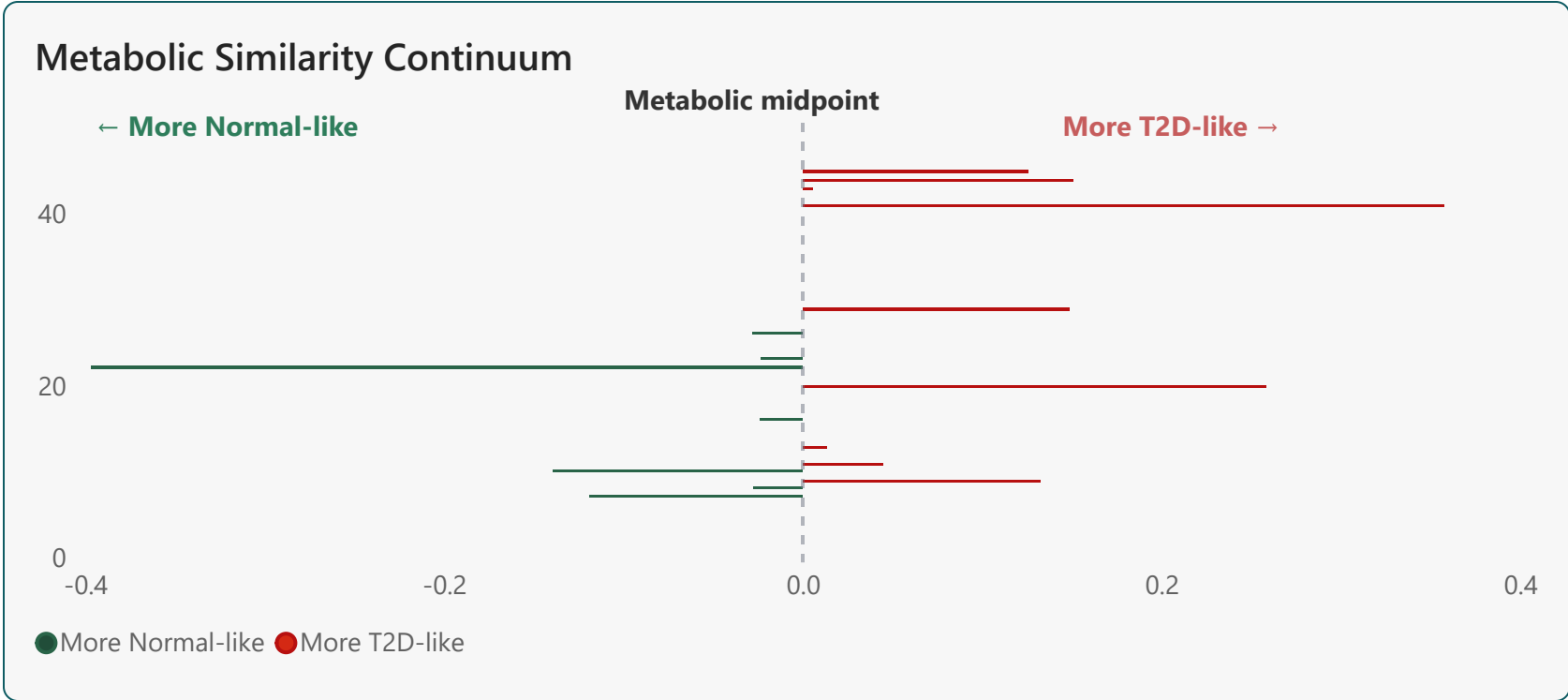
Average glucose and glucose variability rise from Normal to Prediabetes to Type 2 Diabetes.



Key Insight: Average glucose, glucose variability, and time above 180 mg/dL all increased from Normal to Prediabetes to Type 2 Diabetes, showing progressively greater glycemic burden and instability.

Prediabetes Is Not One Uniform State

Participants classified as Prediabetes span a continuum from more Normal-like to more Type 2 Diabetes-like metabolic profiles.



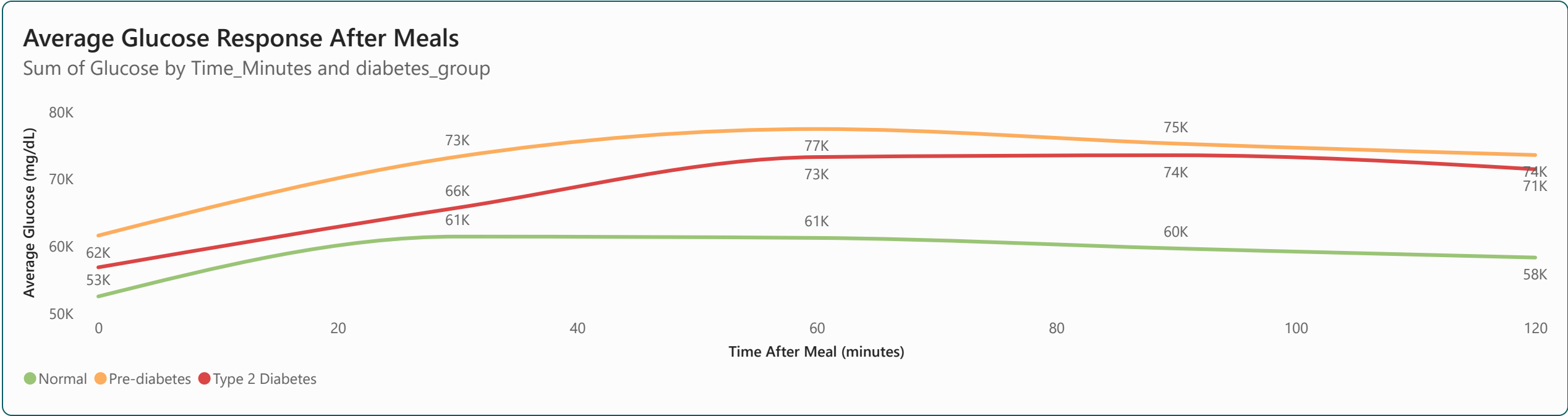
How Different Are the Two Ends of the Continuum?

Left — Normal-like	Right — T2D-like
Avg glucose: 81.81	Avg glucose: 125.56
Time >140: 2.05%	Time>140: 27.49%
Fasting glucose: 105	Fasting glucose: 125.3
TAR >180: 0.16%	TAR>180: 6.26%
BMI: 27.86	BMI: 33.25
TG/HDL: 1.52	TG/HDL: 2.60

Key Insight: *Prediabetes is heterogeneous: participants with the same diagnostic label can show substantially different glucose exposure and metabolic profiles.*

How Does Post-Meal Glucose Response Change Across Metabolic Groups?

From Normal to Prediabetes to Type 2 Diabetes, glucose rises higher, stays elevated longer, and recovers more slowly.



Time to Peak

Avg minutes to peak glucose

**Normal: 56.96 min**

**Prediabetes: 62.31 min**

**T2D: 71.19 min**

Peak Rise card	
Peak – Baseline (mg/dL)	
Normal	Average of peak_rise 28.19
Pre-di...	Average of peak_rise 41.68
Type 2...	Average of peak_rise 58.95

2-hour iAUC	
Normal	Average of iauc_2hr 1.63K
Pre-dia...	Average of iauc_2hr 2.63K
Type 2 ...	Average of iauc_2hr 3.77K

Glucose Above Baseline at 120 min	
Normal	Average of glucose_change_120 9.19
Pre-dia...	Average of glucose_change_120 19.56
Type 2 ...	Average of glucose_change_120 32.10

Key Insight: As metabolic status worsened, meals produced larger glucose spikes, greater 2-hour glucose exposure, and more residual elevation at 120 minutes.

What Information Best Predicts Post-Meal Glucose Response?

Meal composition alone was not enough — prediction improved when person-level metabolic context and activity were added.

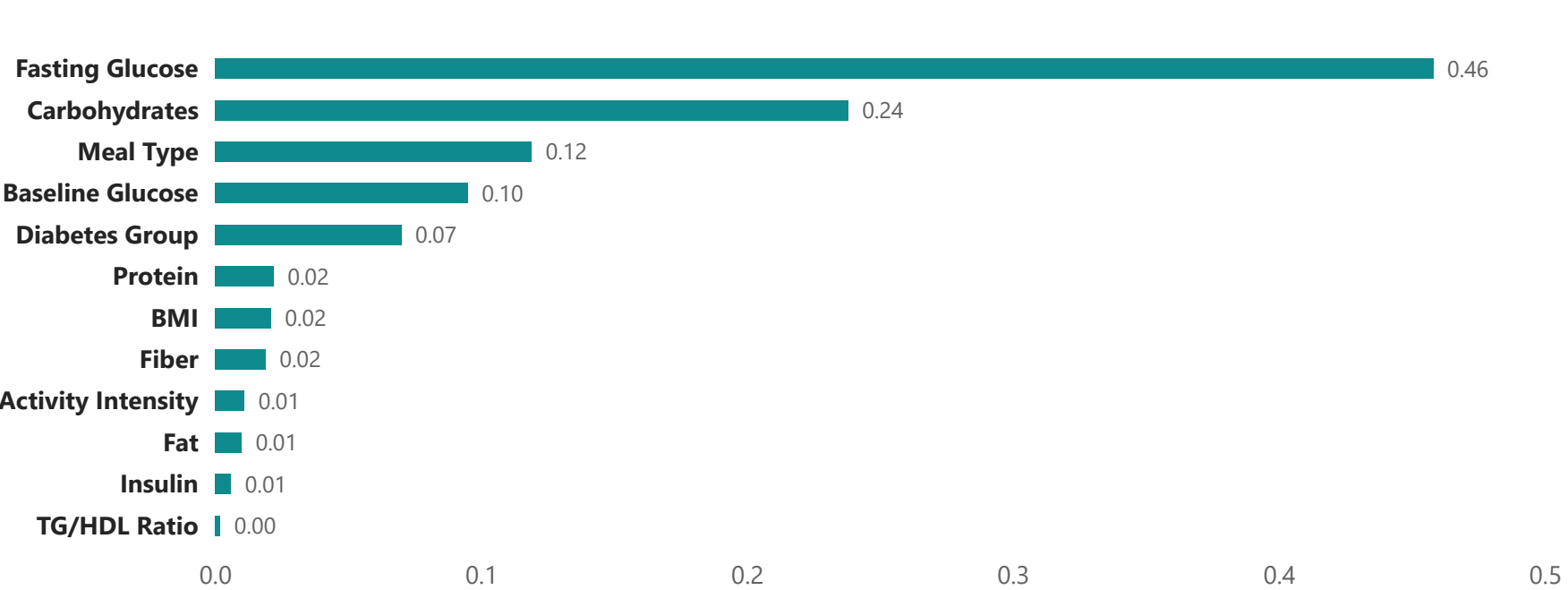
Model Performance Comparison

Higher R² indicates better predictive performance

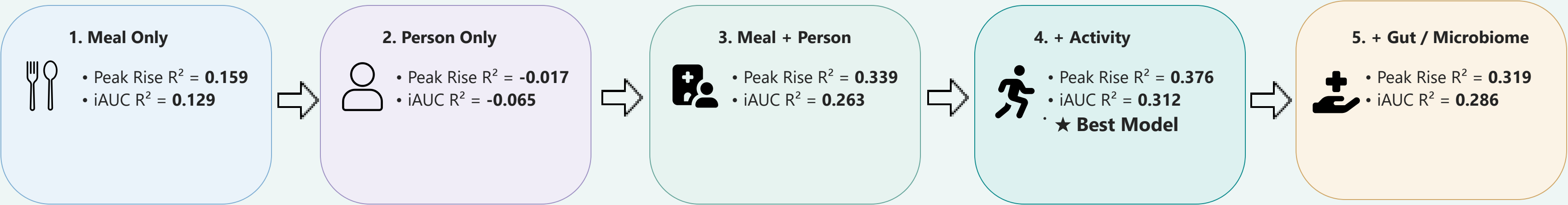


Top Predictive Features

Cross-validated permutation importance



Model Journey

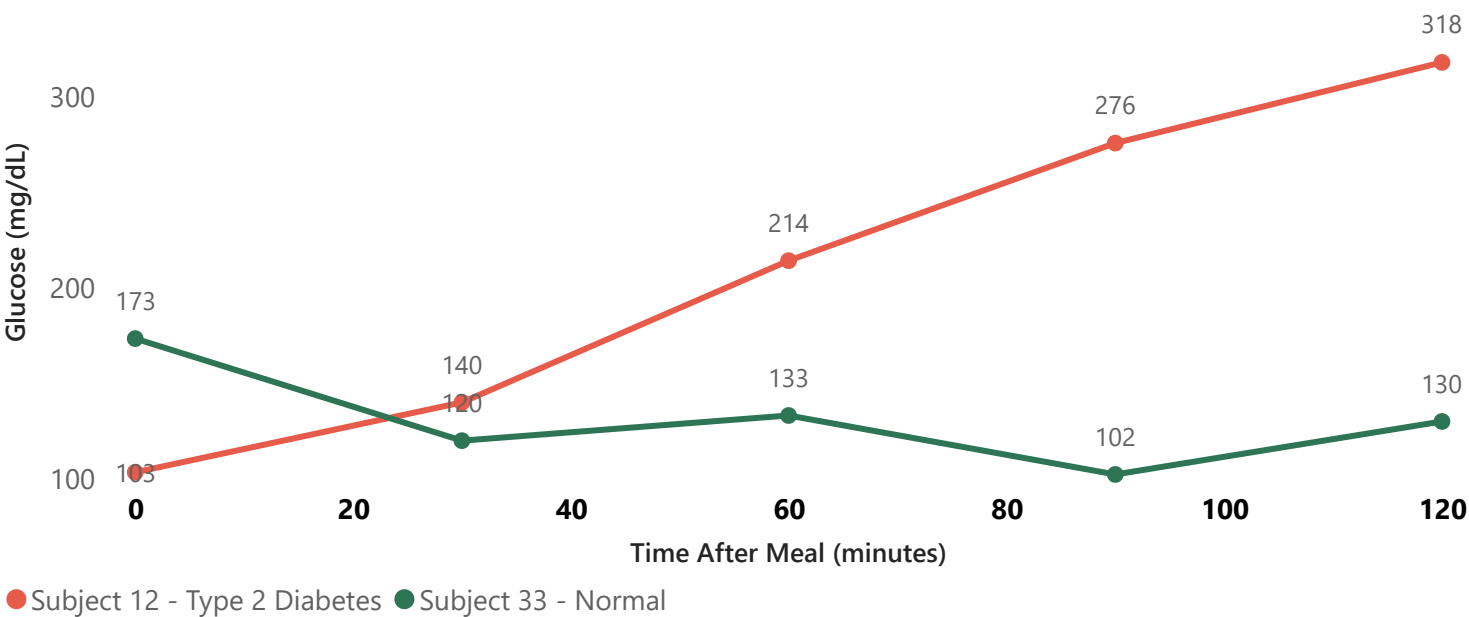


Key Insight: The best prediction came from combining meal composition, individual metabolic profile, and post-meal activity. Gut/microbiome summary features did not add incremental predictive value in this exploratory sample.

Same Recorded Macros. Very Different Glucose Response

Two participants consumed meals with the same recorded macronutrient composition, yet their glucose trajectories were dramatically different.

Post-Meal Glucose Response Over Time



Matched Meal (Lunch)

 **Carbs:** 94 g

 **Protein:** 12 g

 **Fat:** 13 g

 **Amount Consumed:** 100%

Peak difference: ~188 mg/dL

Normal — Subject 33

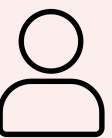


Baseline glucose: 173.47

Peak rise: 0

Fasting glucose: 116

T2D — Subject 12



Baseline glucose: 103.33

Peak glucose: 318.33

Peak rise: 215

Fasting glucose: 179

Key Insight: *The same recorded macronutrient composition produced markedly different glucose trajectories. This supports our broader model finding that meal information alone is insufficient; individual metabolic context is essential for personalized prediction.*